01. REFLEX KLYSTRON MODE CHARACTERISTICS

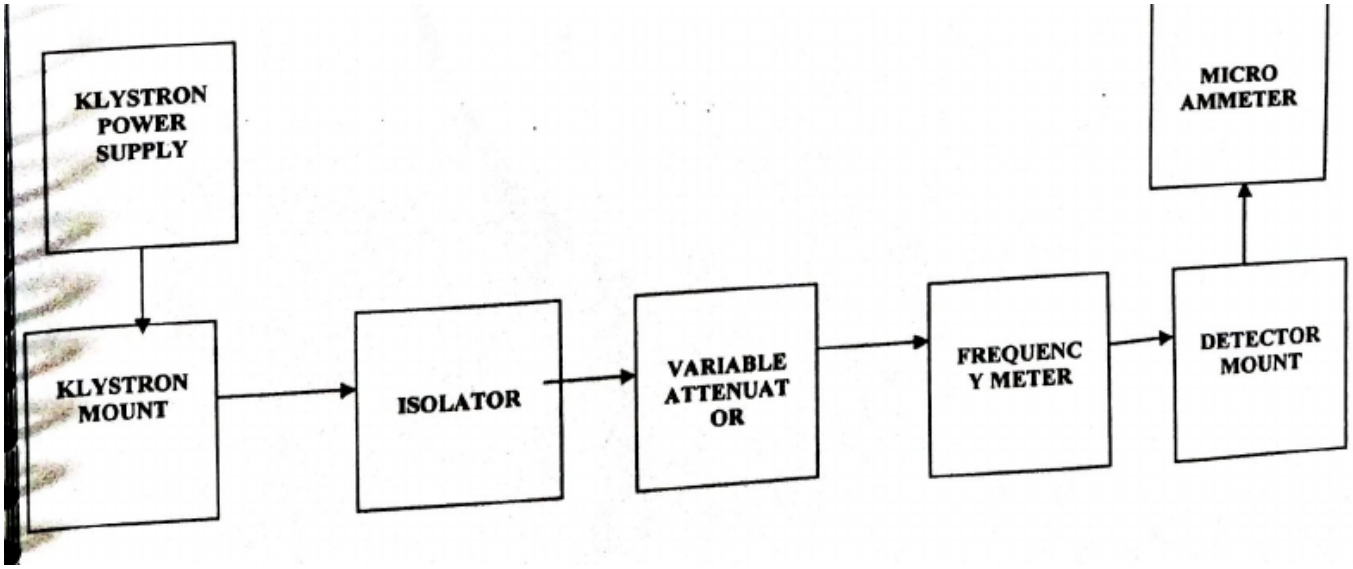
EQUIPMENT REQUIRED:-

Klystron power supply, klystron mount, Isolator, Frequency meter, Variable attenuator, detector mount, Waveguide stand, Ammeter

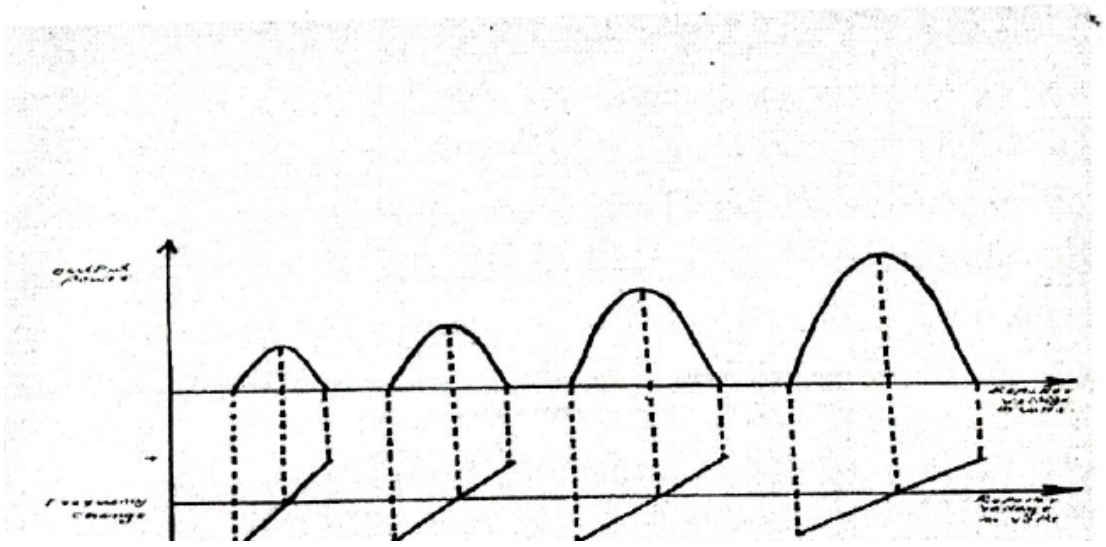
PROCEDURE:-

1. Connect the components and equipments as shown in the figure.
2. Set the variable attenuator is maximum.
3. Set the modulation knob of klystron power supply to AM position, beam voltage knob to full anti clockwise (minimum) and repeller voltage knob to fully clockwise (maximum) and the power switch to OFF position and heater switch(HT) is in OFF position
4. Rotate frequency meter at one side full.
5. Switch on the klystron power supply and cooling fan for klystron tube.
6. Put the HT switch in ON position
7. Observe the meter select knob in the current position, when the current increase at 10mA turn the meter select switch to voltage position and rotate beam voltage knob clockwise slowly up to 300v reading.
8. observe the beam current on the meter by changing the meter switch to current position the current should not increase more than 25mA
9. Change the meter select knob to repeller position and repeller voltage decrease slowly and set the maximum deflection current and take the frequency on the frequency meter.
10. Change the repeller voltage to minimum and read the current and frequency for each repeller voltage at 3 different modes and plot the graph.

FIGURES:-



Block diagram



Waveforms: output power and frequency change vs repeller voltage (volts)

CALCULATIONS / FORMULAE:-

Transit time: $T = (n + 3/4) / F_o$

Mode number: $N_n = (n + 3/4)$; $V_1/V_2 = N_2/N_1$

(V_1, V_2 = repeller voltages needed to operate the klystron in a given mode, for fixed anode voltage and constant frequency)

02. GUNN DIODE CHARACTERISTICS

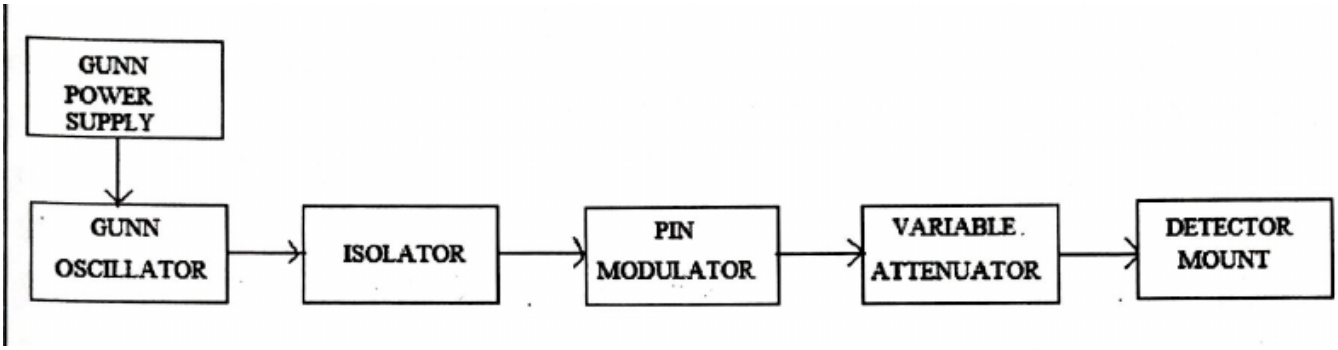
EQUIPMENT REQUIRED:-

Gunn oscillator, Gunn power supply, PIN modulator, isolator, frequency meter, variable attenuator, detector mount, multi meter

PROCEDURE:-

1. Set the components as shown in the block diagram.
2. Keep the control knobs of Gunn power supply as shown below.
   Gunn Bias knob: Fully anticlockwise
   PIN Bias knob: Fully anticlockwise
   PIN Mode Frequency: Any position
1. Set the micrometer of Gunn Oscillator for required frequency of operation
2. Measure the Gunn diode current corresponding to the various Gunn bias voltages through the digital panel meter. Do not exceed the bias voltage above 8 volts
3. Plot the voltage and current readings on the graph as shown in the figure.
4. Measure the threshold voltage with responds to maximum current.

FIGURES:-



Block diagram

CALCULATIONS / FORMULAE:-

No formula is given in the sheet. Plot Gunn bias voltage (V) against Gunn diode current (mA); the threshold voltage is the bias voltage at which the current is maximum (bias not to exceed 8 V).

03. VSWR AND FREQUENCY MEASUREMENT

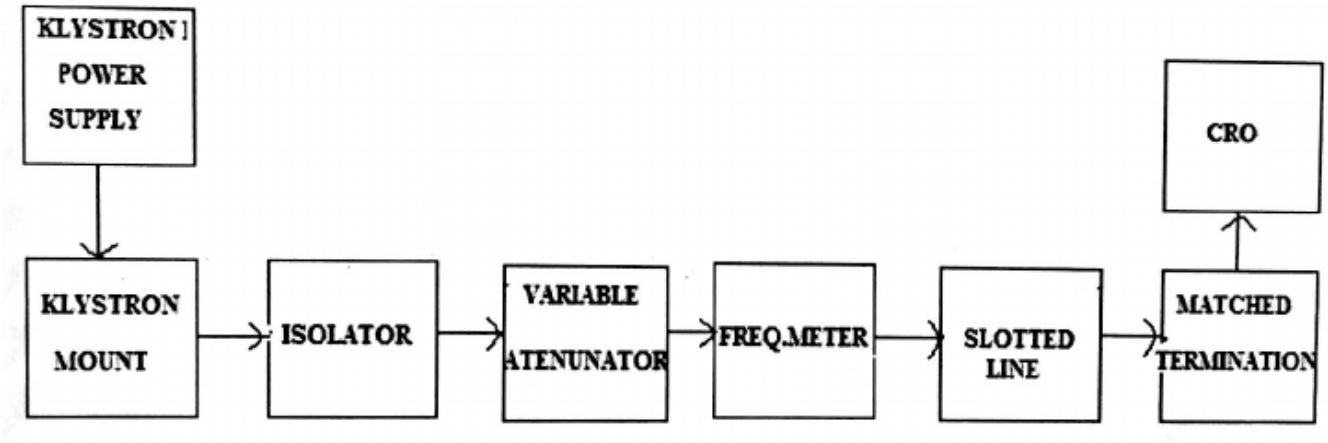
EQUIPMENT REQUIRED:-

Klystron mount, klystron power supply, isolator, frequency meter, variable attenuator, slotted line, matched termination, CRO

PROCEDURE:-

1. Set up the component and equipments as shown in figure.
2. Set the variable attenuator at maximum position
3. Keep the Control knobs of Klystron power supply as below
   - Mod-Switch: AM
   - Beam Voltage knob: Fully antilock wise
   - Repeller Voltage: Fully clockwise
   - AM-Amplitude Knob: Around fully clockwise
   - AM-Frequency: Around Mid position
4. Switch on the klystron power supply, cooling fan, CRO
5. Turn the meter switch of power supply to beam voltage position and set beam Voltage 300v with the help of beam voltage knob and current 10mA
6. Adjust the repeller voltage to get a square wave in CRO.
7. Tune the plunger of Klystron mount for maximum deflection.
8. Move the slotted line to get maximum voltage Emax and minimum voltage On the CRO
9. Calculate the VSWR by using the equation VSWR = Emax/Emin
10. Also calculate the reflection coefficient ρ = S-1/S+1 where S = VSWR of the Te10 mode wave guide.

FIGURES:-



Block diagram

CALCULATIONS / FORMULAE:-

VSWR: $S = V_{max} / V_{min} = E_{max} / E_{min}$ (varies from 1 to ∞; minimum value is unity)
Reflection coefficient: $\rho = (S - 1) / (S + 1)$, S = VSWR of the TE_{10} mode waveguide
Result: VSWR = ... and Reflection Coefficient = ...

04. VERIFY THE RELATION BETWEEN GUIDE WAVELENGTH, FREE SPACE WAVELENGTH AND CUT-OFF WAVELENGTH FOR RECTANGULAR WAVEGUIDE

AIM:-

To determine the frequency and wavelength in a rectangular waveguide working on TE_{10} mode.

EQUIPMENT REQUIRED:-

Klystron tube, isolator, frequency meter, variable attenuator, slotted section, waveguide stand, VSWR meter, movable short and matched termination.

THEORY:-

Modes in a waveguide are represented as TE(m,n) / TM(m,n), where TE = transverse electric, TM = transverse magnetic and m = number of half-wavelength variations in the broader direction. The TE_{10} mode has the lowest cut-off frequency and is hence the dominant mode.

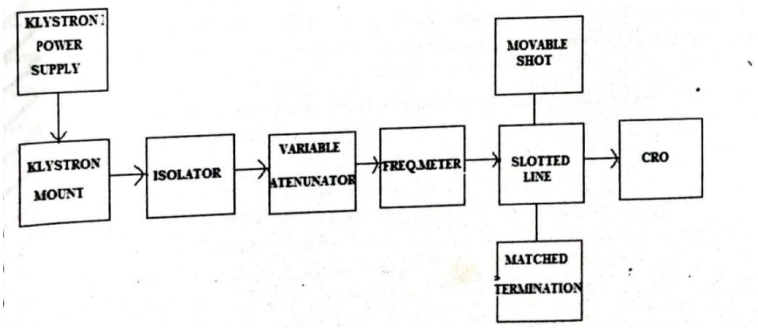
For the dominant TE_{10} mode in a rectangular waveguide, λ_0, λ_g and λ_c are related as: $\mathbf{(1/\lambda_0)^2 = (1/\lambda_g)^2 + (1/\lambda_c)^2}$

where λ_0 = free space wavelength, λ_g = guide wavelength, λ_c = cut-off wavelength. For the TE_{10} mode, $\lambda_c = 2a/m = 2a$ (m = 1), where "a" is the broad dimension of the waveguide. Also $c = f\lambda$, where $c = 3\times10^8$ m/s is the velocity of light and f is the frequency.

PROCEDURE:-

1. Set up the components and equipments as shown in figure.
2. Set the variable attenuator at maximum position.
3. Keep the control knobs of the klystron power supply as below: Mod-Switch: AM; Beam Voltage knob: fully anticlockwise; Reflector Voltage: fully clockwise; AM-Amplitude knob: around fully clockwise; AM-Frequency: around mid position.
4. Switch on the klystron power supply, CRO and cooling fan.
5. Turn the meter switch of the power supply to beam voltage position and set the beam voltage at 300 V with the beam voltage knob, current around 15 to 20 mA.
6. Adjust the reflector voltage to get a square wave in the CRO.
7. Tune the plunger of the klystron mount for maximum deflection.
8. Move the slotted line to get two successive maximum/minimum deflections and take two successive distances d_1 and d_2 and the frequencies.
9. Replace the movable short by the matched termination and detune the frequency meter, and again take d_1 and d_2 and the frequencies.
10. Measure the waveguide inner broad dimension "a", which will be around 22.86 mm for X-band.
11. Calculate the guide wavelength as twice the distance between two successive minima/maxima: $\lambda_g = 2(d_1 - d_2)$.
12. Calculate the frequency by the equation $f = c/\lambda_0 = c\cdot\sqrt{[(1/\lambda_g)^2 + (1/\lambda_c)^2]}$, where $c = 3\times10^8$ m/s.
13. Verify with the frequency obtained by the frequency meter.

BLOCK DIAGRAM:-



OBSERVATION TABLE:-

d1 (cm)	d2 (cm)	λg (cm)	λc (cm)	f calc. (GHz)	f meter (GHz)

RESULT:-

Verified the frequency and wavelength in a rectangular waveguide.

INFERENCE:-

05. ANTENNA RADIATION PATTERN MEASUREMENTS

AIM:-

To measure the radiation pattern and gain of waveguide (horn) antennas.

EQUIPMENT REQUIRED:-

Microwave source (Gunn oscillator / klystron), frequency meter, isolator, variable attenuator, detector mount, antennas (transmitting and receiving horns), SWR meter and accessories

PROCEDURE:-

Antenna radiation pattern plotting

1. Set up the equipments as shown in figure, keeping the axis of both antennas in the same axis line.
2. Energize the microwave source for maximum output at the desired frequency with square wave modulation, by tuning the square wave amplitude and frequency of the modulating signal of the Gunn power supply (about 1 kHz) and tuning the detector.
3. Obtain full scale deflection (0 dB) on the normal dB scale at any convenient range switch position of the VSWR meter, by the gain control knob of the VSWR meter or by the variable attenuator.
4. Turn the receiving horn to the left in 2° or 5° steps up to 40°–50° and note the corresponding VSWR meter dB reading in the normal dB range. When necessary, switch the range to the next higher range and add 10 dB to the observed value.
5. Repeat the above step, but this time turn the receiving horn to the right and note down the readings.
6. Draw the relative power pattern, i.e. output vs angle.
7. From the diagram, determine the 3 dB width (beam width) of the horn antenna.

Gain measurement:

a) Set up the equipments as in figure. Both horns should be in line.

b) Keep the range dB switch of the VSWR meter at 50 dB position with the gain control full.

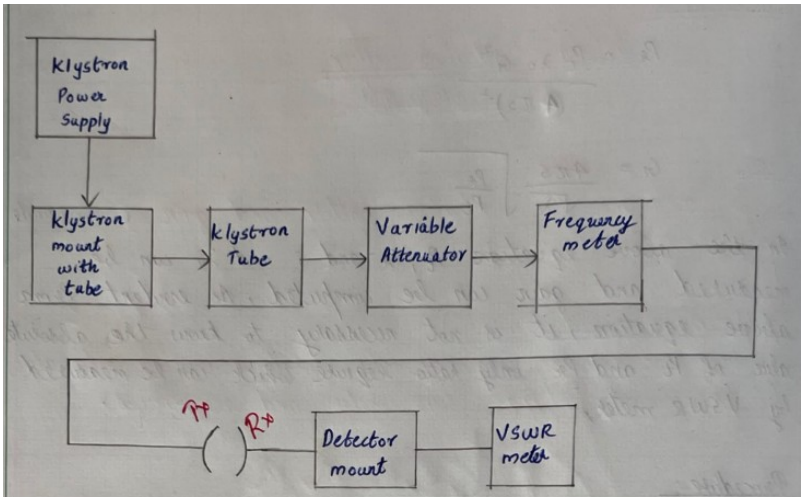
c) Energize the Gunn oscillator for maximum output at 50 dB position with modulating amplitude and frequency of the Gunn power supply and by tuning the detector.

d) Obtain full scale deflection in the VSWR meter with the variable attenuator.

e) Replace the transmitting horn by the detector mount and change the approximate range dB position to get the deflection "on scale". Note and record the range dB position and deflection of the VSWR meter.

f) Calculate the difference in dB between the power measured in the two cases (steps 4 and 5).

FIGURES:-



Block diagram

OBSERVATION TABLE:-

Angle (°)	Left (dB)	Right (dB)
0		
5		
10		
15		
20		
25		
30		
35		
40		
45		

CALCULATIONS / FORMULAE:-

Beam width: angle between the two points at which the power falls to −3 dB of the peak (0 dB) on the radiation pattern.

Gain: $G\ (dB) = \frac{1}{2}\left[(P_r/P_t)\ (dB) + 20 \log_{10}(4\pi R/\lambda) \right]$, where R = distance between the horns and λ = wavelength.

Result: Beam width = ... ° and Gain = ... dB

04. MEASUREMENT OF NUMERICAL APERATURE OF A FIBER, AFTER PREPARING THE FIBER ENDS

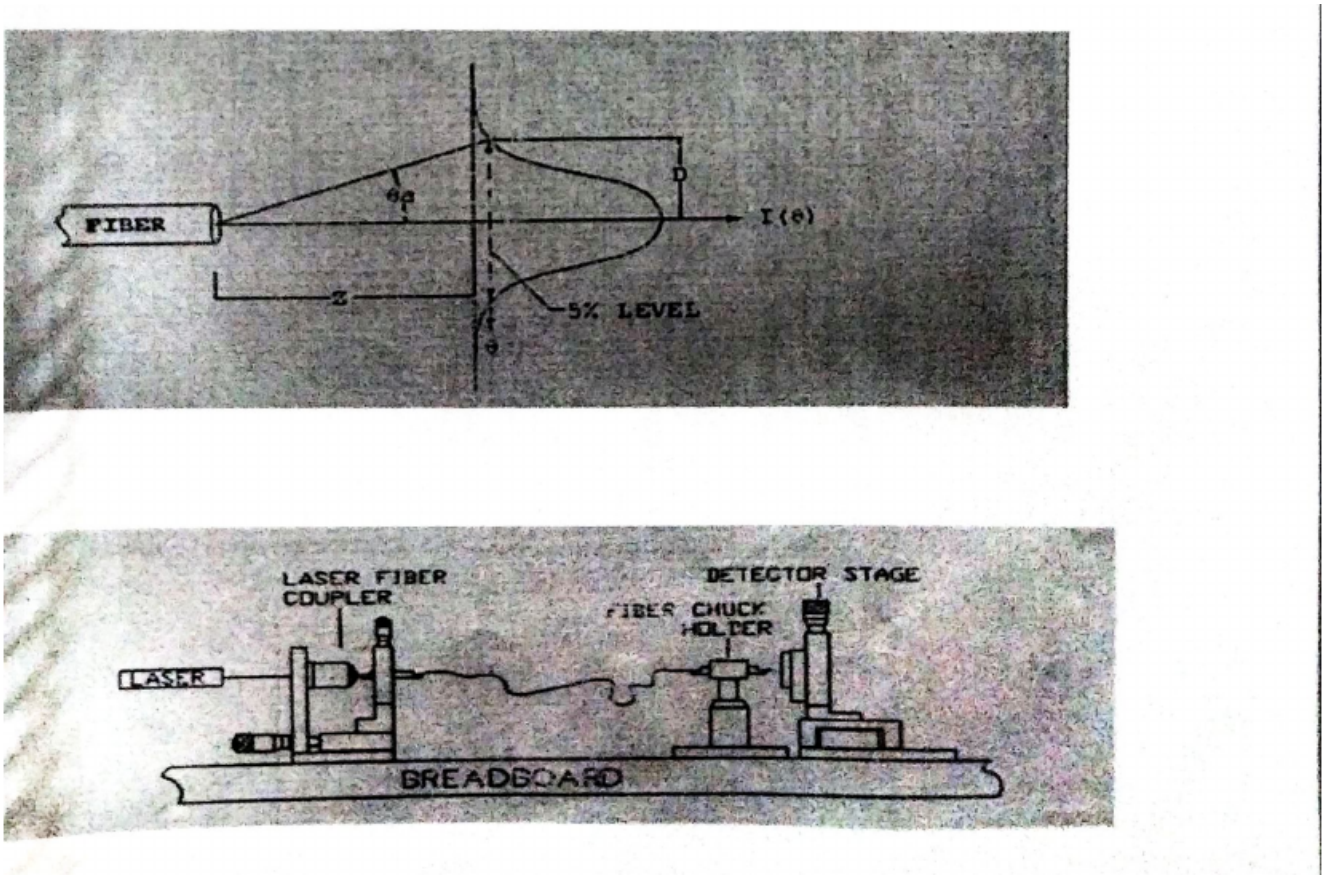
EQUIPMENT REQUIRED:-

Laser diode, Laser-fiber coupler, Fiber chuck holder, Breadboard Detector with translation stage, output measurement unit, Fiber samples.

PROCEDURE:-

a) Mount diode laser to the laser mount.
b) Mount fiber on the Laser-fiber coupler.
C) Collimate the laser beam to the fiber.
d) Fine Adjustment can be done by tilting the fiber postitoner.
e) Fix the output end of fiber to the fiber chuck holder.
f) place pin hole detector nearer to the tip of the fiber output as shown in figure.
g) Take necessary measurements.
h) The experiment can be repeated for various values of 'D' by changing the distance 'Z'.

FIGURES:-



Far-field intensity pattern and experimental setup (laser, laser-fiber coupler, fiber chuck holder, detector stage on breadboard)

CALCULATIONS / FORMULAE:-

Numerical Aperture: NA = sin θ = sin($\tan^{-1}$ D/Z)
D = diameter of the far-field intensity at the 5% intensity level of the maximum intensity; Z = distance between the detector and the fiber output end
Acceptance angle: $\theta_a = \sin^{-1}[NA]$ or $\theta_a = \tan^{-1}(D/Z)$

VOLTAGE VS CURRENT (VI) CHARACTERISTICS OF LED

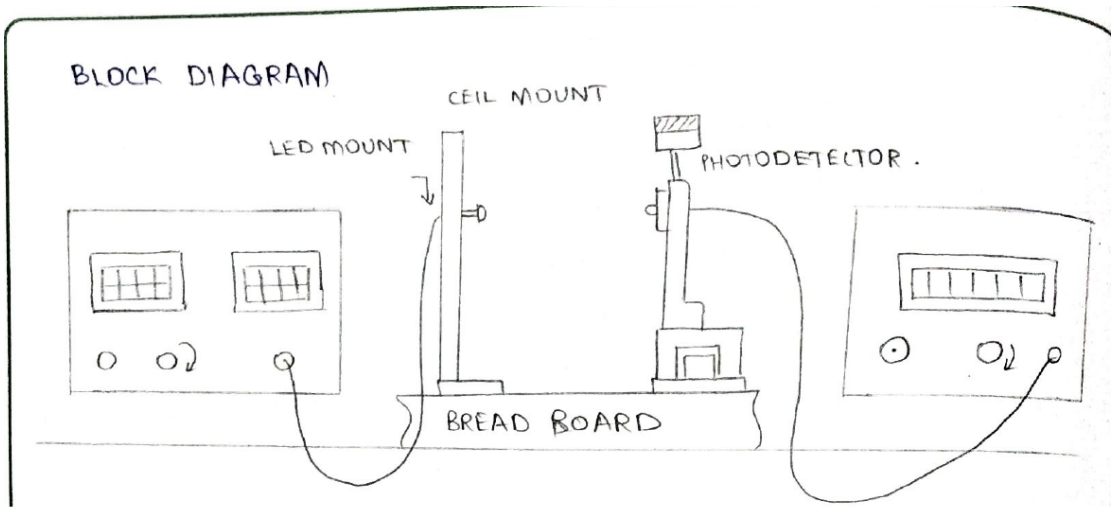
EQUIPMENT REQUIRED:-

Fiber optics Light Emitting Diode module, LED, optoelectronics measurement unit, probes

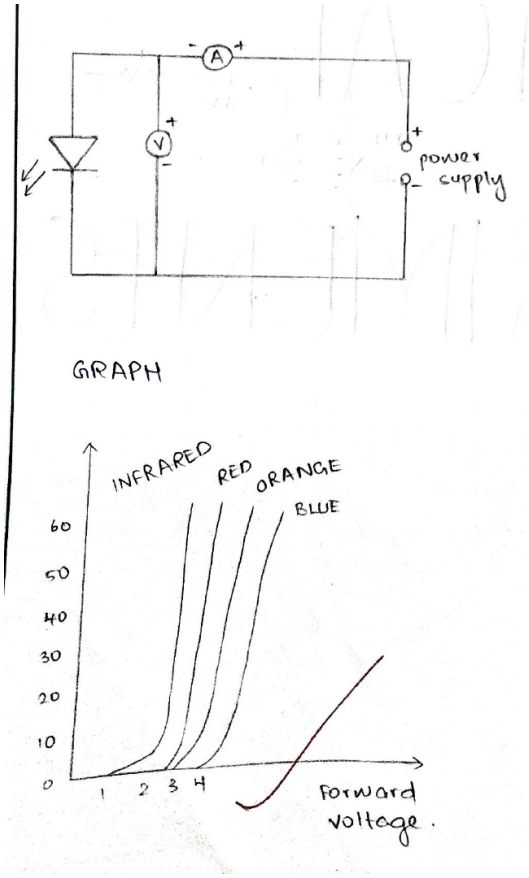
PROCEDURE:-

1) Place an LED on the standard mount.
2) Connect the leads to opto electronics measurement unit
3) switch on the power supply and fix the current at 5MA.
4) Place a photodetector close to the LED to collect almost all the radiant energy.
5) Vary the voltage across the LED at regular intervals of not each time the voltage and corresponding current.
6) Plot the V-I graph.

FIGURES:-



Block diagram



Circuit diagram and V-I graph (infrared, red, orange, blue LEDs; forward voltage on x-axis)

CALCULATIONS / FORMULAE:-

No formula is given in the sheet. Plot forward voltage (V) against current (mA) for the LED.

CHARACTERISTICS OF PHOTODIODE AND MEASURE THE RESPONSIVITY

EQUIPMENT REQUIRED:-

Diode laser, Bread board, lens with mount, Polarizer, photodiode with mount, translation stage, opto-electronic measurement unit.

PROCEDURE:-

Response characteristics

i) Mount diode laser to the laser mount

ii) Place lens with mount infront of laser beam.

iii) Place polarizer rotator such as enlarged laser beam from lens falls on polarizer.

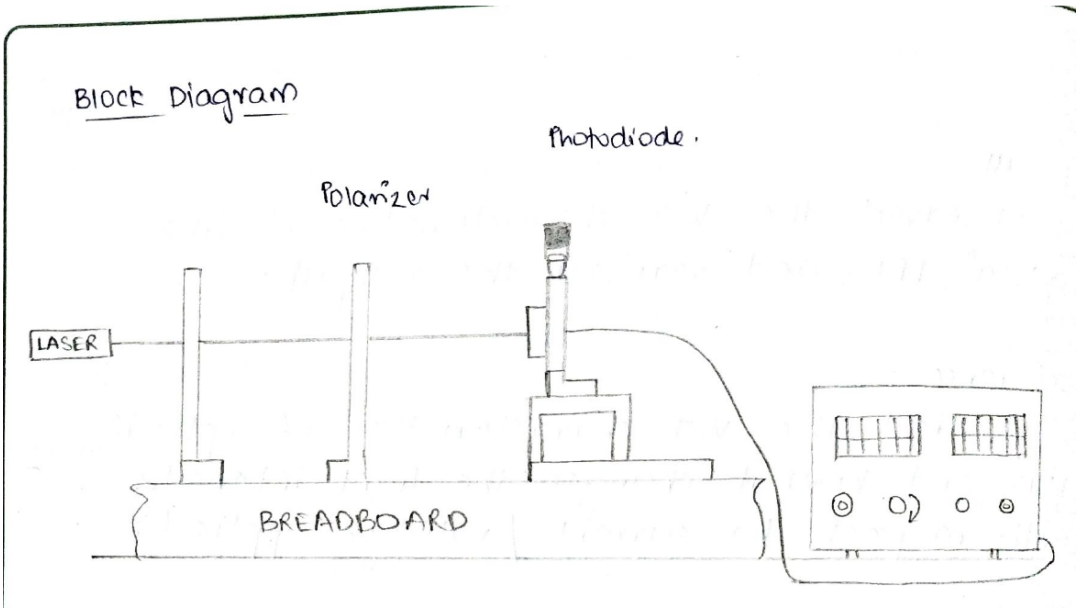
iv) Detatch pinhole photo transistor from its mount (translation stage) and mount photodiode to it

v) connect the photodiode on to the opto-electronic measurement unit

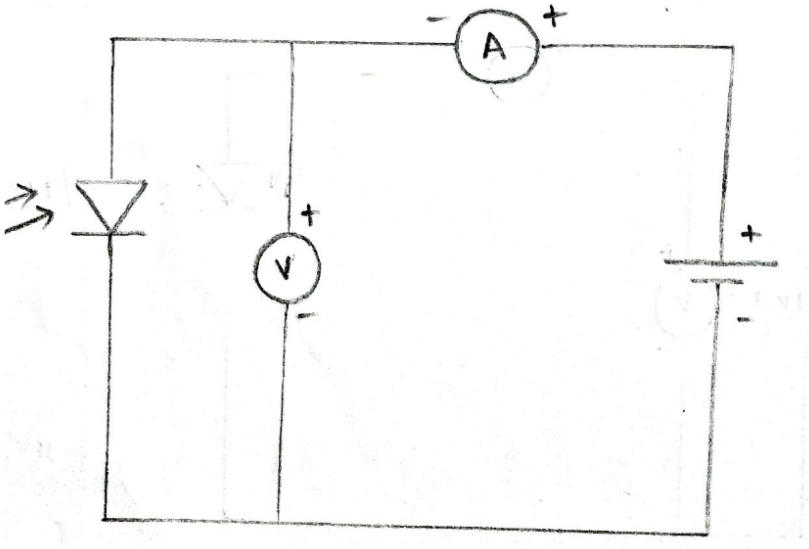
vi) Vary the light intensity by rotating polarizer and note the rotated degree and corresponding output in measurement unit

vii) Repeat the task and plot the graph.

FIGURES:-



Block diagram



Circuit diagram

CALCULATIONS / FORMULAE:-

No formula is given in the sheet. Plot the measurement-unit output against the polarizer rotation angle.

Responsivity (standard definition, not in the sheet): $R = I_p / P$ (A/W) = photocurrent / incident optical power